



**UNIVERSITY
OF ALBERTA**

CMPUT 461 A1 (54551)

Fall 2026

INTRO NATURAL LANGU PROCESSING

Territorial Acknowledgement

The University of Alberta, its buildings, labs and research stations are primarily located on the territory of the Néhiyaw (Cree), Niitsitapi (Blackfoot), Métis, Nakoda (Stoney), Dene, Haudenosaunee (Iroquois) and Anishinaabe (Ojibway/Saulteaux), lands that are now known as part of Treaties 6, 7 and 8 and homeland of the Métis. The University of Alberta respects the sovereignty, lands, histories, languages, knowledge systems and cultures of all First Nations, Métis, and Inuit nations.

Instructor Information

Instructor

Denilson Barbosa

Contact Information

cmput461@ualberta.ca

Office Hours: ONLINE -- see eClass booking information

Lecture Room & Time

see Beartracks

Calendar Description

Natural language processing (NLP) is a subfield of artificial intelligence concerned with the interactions between computers and human languages. This course is an

introduction to NLP, with the emphasis on writing programs to process and analyze texts, covering both foundational aspects and applications of NLP. The course aims at a balance between classical and statistical methods for NLP, including methods based on machine learning. Prerequisites: 201 or 275, and any 300-level Computing Science course.

Additional Course Information (General Content)

The course contrasts classical (pre-LLM) methods with language models in building NLP applications (assistants, chatbots, etc.).

The course does not cover LLM libraries or toolkits, nor does it cover LLM-centered development (e.g., prompt engineering).

Course Objectives/Outcomes

- Pre-processing large natural language datasets containing thousands of documents
- Developing and integrating natural language processing tools for analysis tasks
- Applying computational methods to extract information, classify content, and identify relationships within and between texts
- Examining natural language data challenges through selecting and justifying appropriate cleaning and modeling procedures
- Presenting nuanced interpretations of their work through discussing and reporting
- Comparing classical and statistical algorithms across multiple applications
- Examining how AI assistants process user input to identify intent, retrieve information, and generate responses
- Understanding how large language models are created and tuned and exploring some of the risks of using them

Required Course Materials

Speech and Language Processing (3rd ed. draft)

Link: <https://web.stanford.edu/~jurafsky/slp3/>

Course Material Access

[Course Material Access](#) allows students to access their required course materials through Canvas for the duration of the term. A flat fee per term will appear in the student's Fee Assessment section in Bear Tracks. Digital formats are prioritized, but print materials are provided if digital formats are not available. **This is an automatic opt in program, but students can choose to opt out**, if they decide it is not for them. Some reasons may include:

- the cost of all required course material is less than the program flat fee for the term
- print materials are preferred over digital formats
- access to course material is desired beyond a given term

Students will automatically be opted out if none of their digital or printed course material has a cost associated with it. For more information on how to access your course materials on Canvas, how to opt out, and opt out deadlines, please visit the [Course Material Access](#) website.

Final Letter Grades

University of Alberta Grading Policy

Grades reflect judgments of student achievement made by instructors and must correspond to the associated descriptor. These judgements are based on a combination of absolute achievement and relative performance in a class. Faculties may define acceptable grading practices in their disciplines. Such grading practices must align with the University of Alberta [Assessment and Grading Policy](#) and its procedures, which are available online at the [University of Alberta Policies and Procedures Online \(UAPPOL\)](#) website.

Course Grades Obtained by Undergraduate Students

Descriptor	Letter Grade	Grade Point Value
Excellent	A+	4.0
Excellent	A	4.0

Descriptor	Letter Grade	Grade Point Value
Excellent	A-	3.7
Good	B+	3.3
Good	B	3.0
Good	B-	2.7
Satisfactory	C+	2.3
Satisfactory	C	2.0
Satisfactory	C-	1.7
Poor	D+	1.3
Minimal Pass	D	1.0
Failure	F or F4	0.0
Note: F4 denotes the eligibility of a student to apply for a re-examination of a course.		

Statement of Expectation for AI Use

You can use GenAI tools in this course, within the guidelines specified for each assessment. Follow all directions as provided. If you have any questions/concerns, please ask. AI is prone to fabrication (factual inaccuracies). Review outputs carefully and validate using trusted sources. You are responsible for any errors or omissions the AI tool provides that you fail to identify and resolve. All graded assignments must be accompanied by the [GenAI Use: Acknowledgement and Reflection form](#), whether you have used AI or not.

Important: AI use must be acknowledged transparently. See the U of A Library's [How to Cite AI](#) for standard reference and citation expectations. Failure to acknowledge AI use may be considered cheating and a violation as outlined in the relevant sections of the University of Alberta [Student Academic Integrity Policy](#).

Missed Term Work/Final Exam Due to Non-medical Protected Grounds (e.g., religious beliefs)

When a term assessment or final exam presents a conflict based on [non-medical protected grounds](#), students can register with the Academic Success Centre for accommodations via their [Register for Accommodations](#) website. Students can

review their eligibility and choose the registration process specific for **Accommodations Based on Non-medical Protected Grounds**.

It is imperative that students review the dates of all course assessments upon receipt of the course syllabus, and register **AS SOON AS POSSIBLE** to ensure the timely application of the accommodation. Students who register later in the term may experience unavoidable delays in the processing of the registration, which can affect the accommodation.

Academic Integrity and Student Conduct

The University of Alberta is committed to the highest standards of academic integrity and honesty. Students are expected to be familiar with these standards regarding academic honesty and to uphold the policies of the University in this respect. Students are particularly urged to familiarize themselves with the provisions of the [Student Academic Integrity Policy](#) (on the [University of Alberta Policies and Procedures Online \(UAPPOL\) website](#)) and avoid any behaviour which could potentially result in suspicions of cheating, plagiarism, misrepresentation, and/or unauthorised collaboration. Academic dishonesty is a serious offence and can result in suspension or expulsion from the University.

The University of Alberta is also committed to maintaining a learning environment that fosters the safety, security, and the inherent dignity of each member of the community, ensuring students conduct themselves accordingly by avoiding behaviour stipulated in the [Student Conduct Policy](#) (e.g., discrimination, harassment, physical assault).

All students are expected to consult the [Academic Integrity website](#) for clarification on the various academic offences. All forms of academic dishonesty are unacceptable at the University. Unfamiliarity of the rules, procrastination or personal pressures are not acceptable excuses for committing an offence. Listen to your instructor, be a good person, ask for help when you need it, and do your own work -- this will lead you toward a path to success. Any academic integrity concern in this course will be reported to the College of Natural and Applied Sciences. Suspected cases of non-academic misconduct will be reported to the

Office of Student Success and Experience (formerly the Office of the Dean of Students). The College, Faculty, and Dean of Students are committed to student rights and responsibilities, and adhere to due process and administrative fairness, as outlined in the [Student Academic Integrity Policy](#) and the [Student Conduct Policy](#). Please refer to the policies for details on inappropriate behaviours and possible sanctions.

The College of Natural and Applied Sciences (CNAS) has created an [Academic Integrity for CNAS Students](#) website. Website content includes the importance of academic integrity, examples of academic misconduct & possible sanctions, and the academic misconduct & appeal process. Students can also access this material as an [online, self-directed Canvas course](#) and complete assessments to test their knowledge.

The Office of Student Conduct and Accountability has also made the following Canvas course available for all students: [Academic Citizenship: Integrity and Belonging in the Learning Community](#).

"Integrity is doing the right thing, even when no one is watching" -- C.S. Lewis

On-Line Homework Disclaimer

Online homework is a component of this course and is provided by a third-party company (GitHub). Please be aware that this company will be storing assessment information that may be associated with you. As a way to protect your personal information, you have the choice to create a new identity with GitHub for this course only. You are not required to provide any identifying information to GitHub. If you have concerns about this, please contact the instructor of the course.

University Policy

Withdrawals

See the [University Calendar](#) for the relevant [add/drop deadlines](#) for each term.

Course Outlines

Policy about course outlines can be found in the [Academic Regulations, Evaluation Procedures and Grading section](#) of the University Calendar.

Student Academic Integrity

The University of Alberta is committed to the highest standards of academic integrity and honesty. Students are expected to be familiar with these standards regarding academic honesty and to uphold the policies of the University in this respect. Students are particularly urged to familiarize themselves with the provisions of the Student Academic Integrity Policy and the Student Conduct Policy (on the [University of Alberta Policies and Procedures Online](#) (UAPPOL) website) and avoid any behaviour which could potentially result in suspicions of cheating, plagiarism, misrepresentation of facts and/or participation in an offence. Academic dishonesty is a serious offence and can result in suspension or expulsion from the University.

Recordings

Audio or video recording, digital or otherwise, of lectures, labs, seminars or any other teaching environment by students is allowed only with the prior written consent of the instructor or as a part of an approved accommodation plan. Student or instructor content, digital or otherwise, created and/or used within the context of the course is to be used solely for personal study, and is not to be used or distributed for any other purpose without prior written consent from the content authors.

Accommodations for Students

In accordance with the University of Alberta's [Accommodation Policy](#) and [Discrimination and Harassment Policy](#), accommodation support is available to eligible students who encounter limitations or restrictions to their ability to perform the daily activities necessary to pursue studies at a post-secondary level

due to medical conditions and/or non-medical protected grounds. Accommodations are coordinated through the [Academic Success Centre](#), and students can learn more about eligibility on the [Register for Accommodations website](#).

It is recommended that students apply AS SOON AS POSSIBLE in order to ensure sufficient time to complete accommodation registration and coordination. Students are advised to review and adhere to published deadlines for accommodation approval and for specific accommodation requests (e.g., exam registration submission deadlines). Students who request accommodations less than a month in advance of the academic term for which they require accommodations may experience unavoidable delays or consequences in their academic programs, and may need to consider alternative academic schedules.

Remote Delivery Considerations

Unless the University mandates otherwise, this course will be delivered in person only.

Lectures will not be recorded.

Exams will be in person.

To successfully participate in remote learning in this course, it is recommended that students have access to a computer with an internet connection that can support the tools and technologies the University uses to deliver content, engage with instructors, TAs, and fellow students, and facilitate assessments and examinations. If you encounter difficulty meeting the technology recommendations, please email the Office of Student Success and Experience (sse@ualberta.ca) directly to explore options and support. Please contact the instructor by the [add/drop deadline](#) if you do not have access to the minimum technology recommended.

Student Resources for Remote Learning:

Online learning may be new to you. Please refer to [Technology for Online Learning - For Students](#) to ensure you have the appropriate technology for remote learning.

Remote Proctoring Consideration:

If mandated by the University, we will use a remote proctoring service for tests/exams. Your computer and environment will be monitored during exams.

- You should arrange for a place to write timed exams without interruption.
- Tablets and mobile devices are NOT compatible with remote proctoring services.

Recordings of Synchronous Activities:

- Please note that class times for this course **will NOT be recorded**.

Grade Evaluation

Title	Assessment Type	Weight	Date of Evalu...
Term test 1	Midterm	25	10/29/26
Term test 2	Midterm	25	12/3/26
Final exam	Exam (final)	45	
Project	Other	5	

For courses with a Final Exam, students must verify the date of the Final Exam on Bear Tracks when the Final Exam Schedule is posted.

Grades are unofficial until approved by the Department and/or Faculty offering the

course.

Re-examination

A student who writes the final examination and fails the course may apply for a re-examination. Re-examinations are rarely granted in the Faculty of Science. Re-examinations are governed by university-wide [Academic Regulations](#) and [Faculty of Science Academic Regulations](#). Misrepresentation of facts to gain a re-examination is a serious breach of the [Student Academic Integrity Policy](#).

Access to Past or Representative Evaluative Material

See Canvas.

AI Assisted Grading

This course uses AI-assisted tools—specifically Claude and Qwen—to support parts of the grading process. These tools assist in identifying patterns, enhancing consistency, and the efficiency of grading tasks. By streamlining these processes, more time can be dedicated to higher-value teaching activities, such as providing personalized feedback, developing learning materials, or directly supporting students, while also ensuring timely return of assessments. AI tools may be used to assist in creating and grading all assessments. Regardless of the tool's involvement, final responsibility for all grades rests with the instructor.

Data Use and Privacy:

Student submissions may be processed through these tools to provide grading support. In alignment with the tools terms of use:

- Your work is **not used to train public AI models**.
- Submitted data is typically **retained temporarily** to complete the grading task but is **not stored long-term**.
- No personally identifiable information is intentionally shared with the AI tool.

Your Options:

If you are uncomfortable with your work being processed through AI-assisted tools, you may choose to opt out. To do so, please notify the instructor by email by September 15, 2026. Your decision to opt out will be respected and will not negatively affect your standing in the course.

Feedback Transparency:

Your feedback will always state how the submission was graded:

- **AI-assisted grading** - Provisional marks generated by the tool and confirmed by the Head TA or the instructor.
- **AI-assisted with human re-check** - Initially processed by the tool but redirected for full human review due to a key performance indicator (KPI) trigger (e.g., low confidence, rubric mismatch, or disagreement between AI grading agents).
- **Manually graded** - Assessed entirely by a human grader without AI involvement.

Appeals & Error Handling:

If you believe your grade reflects an oversight—by the AI tool or a human reviewer, please review the rubric and contact me or the Head TA within one week for re-evaluation.

Policies for Late and Missed Term Work

Missed project deliverables:

The project has four parts, and there will be four deliverable dates during the term. You can submit only one part at a time, and you can submit only one part after successfully completing all previous ones. **There will be no extensions, substitutions, or exceptions to the rules above.**

Missed term exam(s):

A student who cannot complete a term exam due to **incapacitating illness**, severe domestic affliction, or other compelling reasons must contact the instructor within two working days of missing the assessment, or as soon as possible, to request an excused absence. If an excused absence is granted, then the student will **write a modified final exam** with all questions in the regular final exam and at least 75%

of the questions from each missed term exam. There will be no extra time added to the modified final exam. **There will be no extensions, exceptions, or substitutions for missed exams.**

An excused absence is a privilege and not a right. There is no guarantee that an absence will be excused. Misrepresentation of facts to gain an excused absence is a serious breach of the [Student Academic Integrity Policy](#). In all cases, instructors may request adequate documentation to substantiate the reason for the absence, at their discretion.

Deferred Final Examination

A student who cannot write the final examination due to incapacitating illness, severe domestic affliction, or other compelling reasons can apply for a deferred final examination. Such an application must be made to the student's home Faculty Office within two working days of the missed exam and must be supported by appropriate documentation or a Statutory Declaration (see calendar on [Attendance](#)). **Deferred exams will take place at a date to be determined, but no sooner than the Winter term break.** Deferred examinations are a privilege and not a right; there is no guarantee that a deferred examination will be granted. The Faculty may deny deferral requests in cases where less than 50% of term work has been completed. Misrepresentation of facts to gain a deferred examination is a serious breach of the [Student Academic Integrity Policy](#).

Contract Cheating and Misuse of University Academic Materials or Other Assets

Contract cheating describes the form of academic dishonesty where students get academic work completed on their behalf, which they submit for academic credit as if they had created it themselves. Contract cheating may or may not involve the payment of a fee to a third party, who then creates the work for the student.

Examples include:

1. Getting someone to write an essay or research paper for you.
2. Getting someone to complete your assignment or exam for you.
3. Posting an essay, assignment, or exam question to a tutorial or study website; the question is answered by a "content expert", then you copy it and submit it as your own answer.
4. Posting your solutions to a tutorial/study website, public server, or group chat and/or copying solutions that were posted to a tutorial/study website, public server, or group chat.
5. Sharing your login credentials to the course management system (e.g., Canvas) and allowing someone else to complete your assignment or exam remotely.
6. Using an artificial intelligence bot or text generator tool to complete your essay, research paper, assignment, or exam solutions for you (without the instructor's permission).
7. Using an online grammar checker to "fix" your essay, research paper, assignment, or exam solutions for you (without the instructor's permission).

Contract cheating companies thrive on making students believe that they cannot succeed without their help; they attempt to convince students that cheating is the only way to succeed.

Uploading the instructor's teaching materials (e.g., course outlines, lecture slides, assignment, or exam questions, etc.) to tutorial, study or note-sharing websites, public servers, or chat apps is a copyright infringement and constitutes the misuse of University academic materials or other assets. Receiving assignment solutions or answers to exam questions from an unauthorized source puts you at risk of receiving inaccurate information.

Appropriate Collaboration

Students need to be able to recognize when they have crossed the line between appropriate collaboration and inappropriate collaboration. If students are unsure, they need to ask instructors to clarify what is allowed and what is not allowed.

Here are some tips to avoid copying on assessments:

1. Do not write down something that you cannot explain to your instructor.
2. When you are helping other students, avoid showing them your work

- directly. Instead, explain your solution verbally. Allowing your work to be copied is also considered inappropriate collaboration.
3. It is also possible that verbally discussing the solution in too much detail may result in written responses that are too similar. Try to keep discussions at a general or higher level.
 4. If you find yourself reading another student's solution, do not write anything down. Once you understand how to solve the problem, remove the other person's work from your sight and then write up the solution to the question yourself. Looking back and forth between someone else's paper and your own paper is almost certainly copying and considered inappropriate collaboration.
 5. If the instructor or TA writes down part of a solution in order to help explain it to you or the class, you cannot copy it and hand it in for credit. Treat it the same way you would treat another student's work with respect to copying, that is, remove the explanation from your sight and then write up the solution yourself.
 6. There is often more than one way to solve a problem. Choose the method that makes the most sense to you rather than the method that other students happen to use. If none of the ideas in your solution are your own, there is a good chance it will be flagged as copying.

Exam Conduct

Please refer to the [Examinations](#) section of the Academic Calendar for more details on Conduct of Exams.

Some key points to be aware of:

- Your student photo ID is required at exams to verify your identity.
- Students must arrive at the specified time to take the exam. Once the exam has started, students must remain in the physical in-person or remote environment for at least 30 minutes. Students who arrive more than 30 minutes late for an in-person exam will not be permitted to take the exam. Students who arrive more than 30 minutes late for an online exam may have their exam attempt removed or disqualified by the instructor. In both cases, students may apply for a deferred examination.
- All cell phones and smartwatches must be turned off and stored in your bags before you enter the exam room.

- Calculators are not allowed in the exam room at any time.
- "Cheat sheets" are not allowed in the exam room at any time.
- If you are found in possession of a "cheat sheet" or any electronic device inside the exam room, you will be reported for investigation of academic integrity misconduct.

Student Supports

Faculty of Science Student Services: The [Faculty of Science Student Services](#) office is located on the main floor of the Centennial Centre for Interdisciplinary Sciences (CCIS). This office can assist with the planning of [Your Academics](#), and provide information related to [Student Life & Engagement](#), [Internship and Careers](#), and [Study Abroad](#) opportunities. Please visit [Advising](#) for more information about what Faculty Academic Advisors can assist you with.

Student Services Directory: The [Student Services Directory](#) helps U of A students find and access support quickly and effectively, bringing together a catalogue of student services from across the university in one convenient location.

Academic Success Centre: The [Academic Success Centre](#) provides professional academic support to help students strengthen their academic skills and achieve their academic goals. Individual advising, appointments, and group workshops are available year round in the areas of Accessibility, Communication, Learning, and Writing Resources. Modest fees may apply for some services.

Writing and Learning Centre: Get support when and where you need it. The centre, located in 2-10 Cameron Library, offers both in-person and virtual supports for undergraduate and graduate students. Strengthen your academic skills by attending a workshop or booking a one-on-one appointment. Their [website](#) provides more information about drop-in appointments, writing groups, Canvas modules and special programming to find what works best for you. Workshops and appointments are free for current U of A students.

Feeling Stressed, Anxious, or Upset? It's normal for us to have different mental

health experiences throughout the year. Know that there are people who want to help. You can reach out to your friends and access a variety of supports available on and off campus at the [Need Help Now](#) webpage or by calling the 24-hour Distress Line: 780-482-4357 (HELP).

Learning and Working Environment

The Faculty of Science is committed to ensuring that all students, faculty and staff are able to work and study in an environment that is safe and free from discrimination, harassment, and violence of any kind. It does not tolerate behaviour that undermines that environment. This includes virtual environments and platforms.

If you are experiencing harassment, discrimination, fraud, theft or any other issue and would like to get confidential advice, please contact any of these campus services:

- [Office of Safe Disclosure & Human Rights](#): A safe, neutral and confidential space to disclose concerns about how the University of Alberta policies, procedures or ethical standards are being applied. They provide strategic advice and referral on matters such as discrimination, harassment, duty to accommodate and wrong-doings. Disclosures can be made in person or online using the [Online Reporting Tool](#).
- [University of Alberta Protective Services](#): Peace officers dedicated to ensuring the safety and security of U of A campuses and community. Staff or students can contact UAPS to make a report if they feel unsafe, threatened, or targeted on campus or by another member of the university community.
- [Office of the Student Ombuds](#): A confidential and free service that strives to ensure that university processes related to students operate as fairly as possible. They offer information, advice, and support to students, faculty, and staff as they deal with academic, discipline, interpersonal, and financial issues related to student programs.
- [Office of Student Success and Experience](#): They can assist students in

navigating services to ensure they receive appropriate and timely resources. For students who are unsure of the support they may need, are concerned about how to access services on campus, or feel like they may need interim support while they wait to access a service, this office is there to help.

Disclaimer

Any typographical errors in this syllabus are subject to change and will be announced in class and/or posted on the course website. The date of final examinations is set by the Registrar and takes precedence over the final examination date reported in the syllabus.

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Dr. Denilson Barbosa, Department of Computing Science, Faculty of Science, University of Alberta (2026).